

Aim: Implementing clustering algorithm (KMeans).

1.Importing the libraries.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
```

Matplotlib is building the font cache; this may take a moment.

2.Creating a simple dataset.

```
data = {
    'X': [1,2,1,5,6,6,7,8,9,8],
    'Y': [1,1,2,6,5,7,6,8,5]
}
```

3.Printing the dataset.

CODE:

```
df = pd.DataFrame(data)
print("Sample Dataset:")
df
```

OUTPUT:

Sample Dataset:

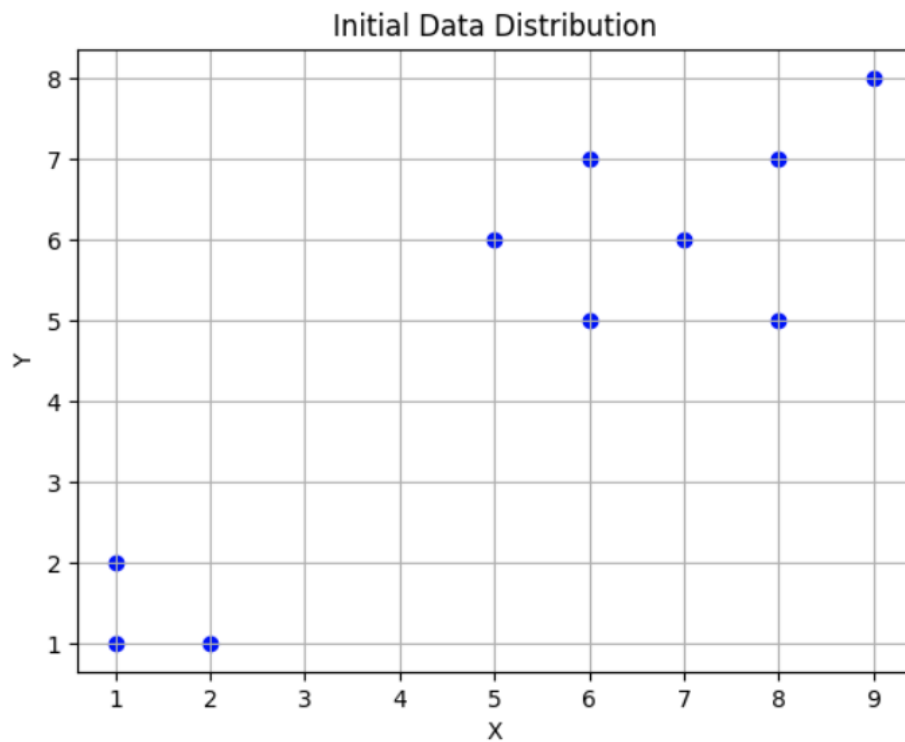
	X	Y
0	1	1
1	2	1
2	1	2
3	5	6
4	6	5
5	6	7
6	7	6
7	8	7
8	9	8
9	8	5

4.Plotting the initial data distribution.

CODE:

```
plt.scatter(df['X'], df['Y'], c = 'blue')
plt.title("Initial Data Distribution")
plt.xlabel("X")
plt.ylabel("Y")
plt.grid(True)
plt.show()
```

OUTPUT:



5. Using KMeans to cluster the data.

CODE:

```
kmeans = KMeans(n_clusters = 2, random_state = 42)
kmeans.fit(df)
df['Cluster'] = kmeans.labels_
print("\nData with Cluster Labels:")
print(df)
```

OUTPUT:

Data with Cluster Labels:

	X	Y	Cluster
0	1	1	0
1	2	1	0
2	1	2	0
3	5	6	1
4	6	5	1
5	6	7	1
6	7	6	1
7	8	7	1
8	9	8	1
9	8	5	1

6. Plotting the result.**CODE:**

```
plt.scatter(df['X'], df['Y'], c = df['Cluster'], cmap = 'viridis')
plt.scatter(kmeans.cluster_centers_[0,0], kmeans.cluster_centers_[0, 1], s = 200, c = 'red', marker = 'X', label = 'Centroids')
plt.title("K-Means Clustering Result")
plt.xlabel("X")
plt.ylabel("Y")
plt.legend()
plt.grid(True)
plt.show()
```

OUTPUT: